GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code: DI05023071

Subject Name: Design for Extrusion Dies

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	MOPEC

Prerequisite:	None
Rationale:	Extrusion is the highest plastic consuming process mainly used for continuous manufacturing of rods, profiles, tubes, pipes, films, sheets, wire and cable etc. Amongst the different items of equipment employed for the operation of extrusion process, the extruder and die are perhaps the most important. Extrusion die design is a complex process that requires careful consideration of various factors to ensure optimal performance, product quality, and cost efficiency. Collaboration between engineers, material scientists, and manufacturing experts is often necessary to achieve the best results. A Plastic Diploma engineer has to supervise the designing and manufacturing process of dies and monitor extrudate production using these dies. This competency requires the knowledge of various designing aspects of extrusion dies. Hence the course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Identify various flows with respect to melt rheology.	R,U
02	Examine extrusion die design factors.	R,U,A
03	Illustrate assembly of die adaptor with screen pack and breaker plate.	R,U,A
04	Design extrusion die for various applications.	R,U,A

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE (E)	PA(M)	PA(I)	ESE(V)	
3	0	0	3	70	30	00	00	100

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
Unit – I Polymer Melt Rheology	1.1 Basic Definitions: shear, shear stress, shear rate & viscosity 1.2 Viscosity under simple shear: Effect of shear stress and shear rate on viscosity of melt 1.3 Velocity profiles of Newtonian and Non-Newtonian fluids 1.4 Types of flow 1.5 Elasticity under simple shear: Visco-elasticity, viscous flow and elastic flow	5	10
Unit- II Basics of Die Design	2.1 Factors affecting die design 2.2 General die design rules 2.3 Materials for extrusion dies 2.4 Equation for output of Newtonian fluid through tubular cross section 2.5 Land length and its importance in die design 2.6 Die geometry 2.7 Die restriction methods and its effect on melt flow 2.8 Die streamlining methods 2.9 Melt fracture phenomenon • Die-swell and its effect on extrudate	13	30
Unit – III Die Adaptor, Breaker Plate & Screen Pack	**Die Adaptor** 3.1 Significance of die adaptor 3.2 Position of die adaptor and its fitting methods 3.3 Factors to be considered for adaptor design **Screen Pack & Breaker Plate** 3.4 Position and functions of screen pack and breaker plate 3.5 Various breaker plate designs • Factors for correct breaker plate assembly	09	20
Unit – IV Types of Extrusion Dies	4.1 Types of various extrusion dies with respect to melt flow direction 4.2 Straight through, crosshead, offset dies and its applications **Flat and Tubular Film Dies** 4.3 Tubular (blown) film dies – side fed & center fed dies 4.4 Constructional features of tubular dies 4.5 Comparison of side fed & center fed tubular dies 4.6 Constructional features of flat film die	18	40

	4.7 Multilayer film dies and its features 4.8 Compare feed block and multi-manifold multilayer dies **Wire Coating Die** 4.9 Pressure and tubing die 4.10 Constructional features of wire coating die and functions of various components **Pipe & Tube Die** 4.11 Constructional features of tube die 4.12 Construction features of straight through and offset pipe die 4.13 Significance of internal and external sizing calibrators **Sheet Dies** 4.14 Constructional features of coat hanger sheet die 4.15 Fish tail die constructional features and its applications **Dies for Solid Sections** • Dies for Rod, tape and profiles		
	Total	**45**	**100**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
25	**50**	**25**	**---------**	**————————**	**————————**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

S.No.	Title of Book	Author	Publication with place, year and ISBN
1	Extrusion of Plastics	Fisher E.G	Plastics and Rubber Institute, 1989, 9780470150122
2	Extrusion Dies for Plastics and Rubbers	Michaeli W., Chapman H.	Carl Hanser Verlag GmbH & Company KG, 2017, 9781569906811

3	Dies for Plastics Extrusion	Joshi M.V.	Macmillan India Limited, 1992, 9780333904497
4	Polymer Extrusion	Rauwendal C.	Carl Hanser Verlag GmbH & Company KG, 2014, 9781569905166
5	Extruding Plastics	Rosato D.V.	Springer US, 2013, 9781461557937
6	Die Design for Extrusion of Plastic Tubes and Pipes	Sushil Kainth	Hanser Publishers, 2018, 9781569906729

(b) Open-source software and website:

1. https://www.researchgate.net/publication/267231878_Basic_Concepts_in_Polymer_Melt_Rheology_and_Their_Importance_in_Processing
2. https://www.dynisco.com/userfiles/files/The_Die_and_Post_Extrusion_Equipment.pdf
3. https://www.gmatw.com/en/article/Here-are-10-key-points-to-understand-extrusion-dies.html
4. https://youtu.be/WaB-dsB1Kfk
5. https://www.youtube.com/watch?v=XfJ5f3FGAuI
6. https://www.youtube.com/watch?v=6OdSNuqeF4E
7. https://www.youtube.com/watch?v=HXZyAz9mMKA&pp=ygUfc2l6aW5nIGNhbGlicmF0b3JzIGZvciBleHRydWRlcg%3D%3D
8. https://youtu.be/zcBv_JvFDBI
9. https://www.youtube.com/watch?v=HXZyAz9mMKA

Suggested Activities for Students:

1. Students will collect information about recent trends in extrusion die design.
2. Students will prepare banners showing constructional features of various dies.
3. Students will visit nearby industries and collect information about die design.

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